

APML Connect Pro

Full Technology Stack & System Architecture Specification

CLIENT & INVESTOR EDITION
Version 2.4 | Enterprise EMR

1. Executive Technical Overview

APML Connect Pro is a high-performance, enterprise-grade Electronic Medical Records (EMR) & Clinical Management System built with a hybrid **Offline-First Mesh Architecture**. It guarantees 100% continuous operational uptime across Reception, Doctor Consultation, and Pharmacy workstations even during total internet outages.

2. Technology Stack Breakdown

Layer	Technologies Used	Key Advantages
Frontend UI Framework	Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS	Server-Side Rendering (SSR), type safety, sub-millisecond route transitions
Design System	Liquid UI (Fluid Glassmorphism, Dynamic CSS Canvas, Blur Orbs)	State-of-the-art modern visual aesthetic, light & dark theme adaptivity
Backend Services	Node.js, Express.js REST API, WebSockets (WS), C# Native Service	Real-time bidirectional synchronization, zero-latency RPC calls
Database & Persistence	Prisma ORM, PostgreSQL (Production) / SQLite (Local Native)	ACID compliance, schema validation, local database replication
Offline Mesh Sync	Autonomous Peer-to-Peer `meshSync` Protocol, LocalStorage Cache	Zero cloud dependency, auto-reconciliation when server reconnects
Mobile & Android App	Native Android Studio Java/Kotlin APK, Progressive Web App (PWA)	1-Tap home screen app installation on Android, tablet & mobile optimized
Security & Compliance	AES-256 Local Encryption, JWT Authentication, Role-Based Control (RBAC)	HIPAA & GDPR audit readiness, isolated role permissions

3. Core Workstation Modules

■ Receptionist Desk (Port 3001)

Patient registration, token generation, check-in queue management, drag-and-drop lab report uploader, billing & invoice printing.

■ Doctor Cabinet (Port 3002)

EHR medical records, instant patient name search, AI diagnostic suggestions, digital prescriptions & vitals tracking.

■ Pharmacy Counter (Port 3003)

Real-time prescription queue, inventory stock management, medicine dispensing & invoice checkout.

4. Key Highlights for Investors & Enterprise Clients

- **Zero Infrastructure Lock-in:** Can be deployed as a native local server without cloud internet or containerized with Docker.
- **Scalable Microservices Architecture:** Decoupled modular architecture allows independent scaling of individual clinical desks.
- **Instant ROI & Low TCO:** Runs efficiently on existing clinic hardware, eliminating monthly recurring cloud SaaS fees.